

Lab 34 — Linux Command Line Essentials on Killercodea

Course: CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)

Topic 06: Operating Systems — Core 2, 28% of Core 2

Exam objective: Use the Linux commands named in Core 2 objective 1.11 for navigation, file management, searching, process control and package management (Core 2 objective 1.11).

Goal

You work every Linux command the A+ objectives name, on a real Ubuntu machine in the browser. Each command is run against real files and real output so the behaviour, not just the syntax, is what you take away.

What you'll produce

A verified Linux command reference with real output recorded for each command in the objective list.

Tools and equipment

Killercoda Ubuntu Playground (<https://killercoda.com/playgrounds/scenario/ubuntu>)

Browser tools used in this lab

- **Killercoda Ubuntu Playground** — <https://killercoda.com/playgrounds/scenario/ubuntu>

Killercoda Ubuntu Playground — a real Linux machine in the browser

Terminal — root@controlplane

```
$ cat /etc/os-release | grep PRETTY
PRETTY_NAME="Ubuntu 22.04.3 LTS"
$ apt-get update -qq && apt-get install -y net-tools dnsutils
Reading package lists... Done
$ ip -brief addr show
lo        UNKNOWN  127.0.0.1/8
eth0      UP        10.244.0.14/24
$ dig +short www.tertiarycourses.com.sg
104.22.35.191
$ chmod 640 ~/applus/logs/system.log && ls -l ~/applus/logs/
-rw-r----- 1 root root 118 Aug 19 09:14 system.log
$ grep -c 'ERROR' ~/applus/logs/system.log
2
$ _
```

- 1 Open the URL and wait for the prompt. You get root on a real Ubuntu machine — nothing to install.
- 2 Start every lab with apt-get update, then install only the packages that lab needs.
- 3 The playground RESETS when idle. If your terminal disappears, reopen it and re-run the setup steps.

Killercoda Ubuntu Playground — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open <https://killercoda.com/playgrounds/scenario/ubuntu> and wait for the terminal. Confirm where you are and who you are running as.

```
pwd && whoami && id
```

2. Install the packages this lab uses and note that apt is the package manager on Debian and Ubuntu.

```
apt-get update -qq && apt-get install -y tree htop nano less
```

3. Create a working directory tree and confirm its structure visually.

```
mkdir -p ~/aplust/{docs,logs,scripts} && tree ~/aplust
```

4. Create files with content so the later commands have real data to operate on.

```
echo 'ERROR: disk full on /dev/sda1' > ~/aplust/logs/system.log && echo 'INFO: service started' >> ~/aplust/logs/system.log && echo 'ERROR: network timeout' >> ~/aplust/logs/system.log
```

5. List files with the long format showing permissions, ownership, size and modification time, including hidden files.

```
ls -la ~/aplust/logs
```

6. Display file contents with cat, then page through a longer file with less to see how each is used.

```
cat ~/aplust/logs/system.log && ls -la /etc | less -E -X | head -20
```

7. Search file contents with grep, which is the command you use to find an error in a log.

```
grep -n 'ERROR' ~/aplust/logs/system.log && grep -c 'ERROR' ~/aplust/logs/system.log
```

8. Find files by name and by type across a directory tree.

```
find /etc -name '*.conf' -type f 2>/dev/null | head -10
```

9. Copy, move and rename files, then confirm the result of each operation.

```
cp ~/aplust/logs/system.log ~/aplust/docs/backup.log && mv ~/aplust/docs/backup.log ~/aplust/docs/archived.log && ls -l ~/aplust/docs
```

10. Check storage and memory usage, which are the first two things to check on a Linux machine reported as slow.

```
df -h && du -sh ~/aplust && free -h
```

11. List running processes and identify the top consumers, then record how you would terminate a runaway process.

```
ps aux --sort=-%mem | head -8 && echo 'Terminate with: kill <PID> or kill -9 <PID>'
```

12. Change file permissions and ownership, then verify both, and record the numeric meaning of the mode you set.

```
chmod 640 ~/aplust/logs/system.log && ls -l ~/aplust/logs/system.log && echo '640 = owner rw-, group r--, others ---'
```

Test it — verification

Every command runs without error on the playground; grep returns exactly two ERROR lines from the log; ls -l confirms the 640 permission as rw-r-----; and your reference records real output for each command rather than expected output.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the <code>apt-get install</code> step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against <code>labs/tools.md</code> . All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete `worksheet.md` as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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